

Question # 2 of 10 (**Start time: 01:52:51 PM, 04 July 2025**)

We can add the matrices of _____.

Select the correct option

☐	same number of columns
☒	same order
☐	different order
☐	same number of rows.

Question # 1 of 10 (Start time: 01:51:54 PM, 04 July 2025)

Which one of the following is a null matrix?

Select the correct option

☐	$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$
☒	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Question # 8 of 10 (Start time: 01:54:29 PM, 04 July 2025)

In LU-decomposition, if A is the square matrix of order m, then the order of both lower and upper triangular matrices will be -----.

Select the correct option

☐	$2m$
☐	$m + 1$
☒	m
☐	m^2

Question # 3 of 10 (Start time: 01:53:10 PM, 04 July 2025)

Which of the following is the most suitable option while doing LU-factorization?

Select the correct option

☐	Columns can be interchanged.
☒	Rows can be interchanged.
☐	Rows cannot be interchanged.
☐	Entries cannot be interchanged.

Question # 6 of 10 (Start time: 01:54:04 PM, 04 July 2025)

If $A = \text{transpose of } A$ (where A is a square matrix), then which of the following is the most appropriate option for A ?

Select the correct option

☒	A is symmetric matrix.
☐	A is singular matrix.
☐	A is invertible matrix.
☐	A is a scalar matrix.

Question # 4 of 10 (Start time: 01:53:25 PM, 04 July 2025)

A matrix in which each descending diagonal from left to right is constant is called a _____ matrix.

Select the correct option

☐	diagonal
☐	block upper triangular
☒	diagonal-constant
☐	block lower triangular

Question # 5 of 10 (Start time: 01:53:47 PM, 04 July 2025)

If $AB = I = BA$ for matrices A , B and I , where I is an identity matrix, then

Select the correct option

☐	A is inverse of B ,
☒	All of the above.
☐	B is inverse of A ,
☐	$A^{(-1)} = B, B^{(-1)} = A$,

Question # 10 of 10 (Start time: 01:55:03 PM, 04 July 2025)

Which of the following is true for the matrix

$$M = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ O & M_{22} & M_{23} \\ O & O & M_{33} \end{bmatrix};$$

where M_{11} , M_{22} and M_{33} are square sub-matrices, and O is a zero sub-matrix?

Select the correct option

☐	It is a Null matrix.
☐	It is diagonal-constant matrix.
☐	It is a block lower triangular matrix.
☒	It is a block upper triangular matrix.*

Question # 9 of 10 (Start time: 01:54:43 PM, 04 July 2025)

If $X = \begin{bmatrix} M \\ N \end{bmatrix}$ and $Y = \begin{bmatrix} Q & P \end{bmatrix}$ (where M , N , Q and P are square sub-matrices of same size), then which of the following is possible ?

Select the correct option

☐	The product XY and YX both are not defined.
☐	The product XY is not defined but YX is defined.
☐	The product XY and YX both are defined.
☒	The product XY is defined but YX is not defined.

Question # 7 of 10 (Start time: 01:54:17 PM, 04 July 2025)

Total Marks: 1

Let A be the matrix of order 2×3 and B be the matrix of order 3×5 , then which of the following is the order of the matrix AB?

Select the correct option

 Reload Math Equations

☐	3×3
☐	3×5
☐	2×3
☒	2×5

Click to Save Answer & Move to Next Question